6th Grade Accelerated Unit 15 Assessment Guide

General Information	Unit 15 builds upon the support for building students' number sense from throughout the course by having students use different forms of a rational numbers in the setting of probability. The benchmarks MA.6.NSO.1.1 (Extend previous understanding of numbers to define rational numbers. Plot, order and compare rational numbers.) and MA.6.NSO.3.5 (Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals and percentages.) are built upon in this unit as students consider the likelihood of an event occurring and how to represent probabilities in different forms. The general nature of probability encourages students to relate decimals, fractions, and percentages, which is why this culminating unit builds upon previously learned content in Number Sense and Operations.
Calculator Information	The questions on this assessment are calculator neutral. It is suggested that the teacher does not allow a calculator.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1a	Benchmark(s)	MA.7.DP.2.1
{Pitcher, Batter}		Recommended Scoring (6 Total Points)	No partial credit should be given.
Question	**1b**	**Benchmark(s)**	MA.7.DP.2.1
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17}		**Recommended Scoring** **(6 Total Points)**	No partial credit should be given.
Question	**2a**	**Benchmark(s)**	MA.7.DP.2.3
10%		**Recommended Scoring** **(8 Total Points)**	No partial credit should be given.
Question	**2b**	**Benchmark(s)**	MA.7.DP.2.3
70%		**Recommended Scoring** **(8 Total Points)**	No partial credit should be given.

<table>
<tr><th>Question</th><th>3</th><th>Benchmark(s)</th><td>MA.7.DP.2.2</td></tr>
<tr><td colspan="2">
<table>
<tr><th>Least to Most Likely to Occur</th><td>$\frac{8}{11}$, 75%, 0.77, 9 out of 10</td></tr>
</table>
</td><th>Recommended Scoring

(10 Total Points)</th><td>No partial credit should be given.</td></tr>
<tr><th>Question</th><th>4</th><th>Benchmark(s)</th><td>MA.7.DP.2.4</td></tr>
<tr><td colspan="2">
<table>
<tr><td></td><th colspan="3">Simulations</th></tr>
<tr><td></td><td>A deck of 16 cards with 16 different letters.</td><td>A fair spinner with 7 equal sections where each is a different color.</td><td>A 10-sided die with the numbers 1 through 10.</td></tr>
<tr><td>A baseball player with an on-base percentage of 70%.</td><td>☐</td><td>☐</td><td>☒</td></tr>
<tr><td>A basketball player making $\frac{5}{7}$ free throws.</td><td>☐</td><td>☒</td><td>☐</td></tr>
<tr><td>A soccer goalie who stops $\frac{1}{2}$ of shots on the goal.</td><td>☒</td><td>☐</td><td>☒</td></tr>
<tr><td>A hockey player with a shooting percentage of 25%.</td><td>☒</td><td>☐</td><td>☐</td></tr>
</table>
</td><th>Recommended Scoring

(15 Total Points)</th><td>Consider giving students 3 points for each correct selection. Consider deducting 2 points for each incorrect selection.</td></tr>
</table>

<table>
<tr><th>Question</th><th>5</th><th>Benchmark(s)</th><th>MA.7.DP.2.4</th></tr>
<tr><td colspan="2">
To use the
o 10-sided die
o deck of 16 cards
o fair spinner with 7 equal sections
to simulate the
o soccer
o hockey
o baseball
o basketball
context 5
o cards
o numbers
o equal sections
should be assigned to
o being on base.
o making a free throw.
o stopping a shot on goal.
o shooting a goal in hockey.

OR

To use the
o 10-sided die
o deck of 16 cards
o fair spinner with 7 equal sections
to simulate the
o soccer
o hockey
o baseball
o basketball
context 5
o cards
o numbers
o equal sections
should be assigned to
o being on base.
o making a free throw.
o stopping a shot on goal.
o shooting a goal in hockey.
</td><td>Recommended Scoring

(6 Total Points)</td><td>No partial credit should be given. A student should have a correctly completed statement to receive all of the points.</td></tr>
</table>

Question	6a	Benchmark(s)	MA.7.DP.2.3
$\frac{13}{21}$		**Recommended Scoring** **(8 Total Points)**	No partial credit should be given.
Question	**6b**	**Benchmark(s)**	MA.7.DP.2.3
$\frac{18}{21}$ or $\frac{6}{7}$		**Recommended Scoring** **(8 Total Points)**	No partial credit should be given.
Question	**6c**	**Benchmark(s)**	MA.7.DP.2.3
$\frac{2}{21}$		**Recommended Scoring** **(8 Total Points)**	No partial credit should be given.

Question	**6d**	**Benchmark(s)**	MA.7.DP.2.2
The event that is less likely to occur is ○ drawing a marble with the letter N from the jar. ○ drawing a marble with the letter R or K from the jar. ○ drawing a marble with the letter other than H from the jar. The event that is most likely to occur is ○ drawing a marble with the letter N from the jar. ○ drawing a marble with the letter R or K from the jar. ○ drawing a marble with the letter other than H from the jar. The probability of the less likely event is closest to ○ 0. ○ 1.		**Recommended Scoring** **(11 Total Points)**	Consider giving students 4 points each for the first two statements and 3 points for the last statement.
Question	**7**	**Benchmark(s)**	MA.7.DP.2.4
The number of spins in ○ Aaliyah's ○ Xavier's simulation results in a(n) ○ experimental ○ theoretical probability that is closer to the ○ experimental ○ theoretical probability.		**Recommended Scoring** **(6 Total Points)**	No partial credit should be given.